

CYPRUS INTERNATIONAL UNIVERSITY

CMPE455 – Embedded Systems

SMART GREENHOUSE SYSTEM USING ARDUINO UNO

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INTRODUCTION



What is a Smart Greenhouse?

A technologically enhanced growing environment where embedded systems automatically monitor and control temperature, humidity, soil moisture, and lighting conditions.



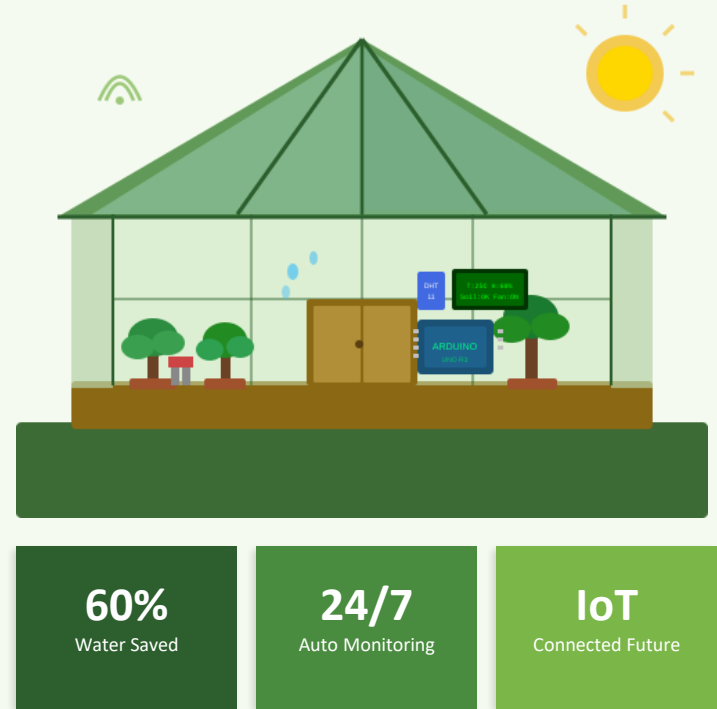
Role of Embedded Systems

Microcontrollers like the Arduino Uno process real-time sensor data and trigger automated responses — enabling smart, responsive agricultural control.



Problem Solved

Eliminates manual monitoring, reduces water waste, prevents crop failure from undetected climate shifts, and lowers operational costs for farmers.



PROJECT OBJECTIVES



Monitor Temperature & Humidity

DHT11 sensor reads ambient conditions continuously

1



Monitor Soil Moisture

Capacitive soil sensor detects irrigation needs

2



Detect Light Intensity

LDR sensor measures ambient light levels

3



Automate Irrigation

Pump activates automatically when soil is dry

4



Control Ventilation

Fan triggers when temperature exceeds threshold

5



Display Live Sensor Data

16×2 I2C LCD shows real-time readings

6

COMPONENTS USED



Arduino Uno

14 digital I/O pins



DHT11 Sensor

Temp & Humidity, $\pm 2^{\circ}\text{C}$ accuracy



Soil Sensor

Analog moisture detection, A0



LDR Sensor

Light-dependent resistor, A1



2CH Relay

Controls pump & fan, 5V trigger



DC Water Pump

12V irrigation pump, D8 relay



DC Fan

12V ventilation fan, D9 relay



16x2 I2C LCD

Display readings, A4/A5 (I2C)



Breadboard

830-point prototype board



Jumper Wires

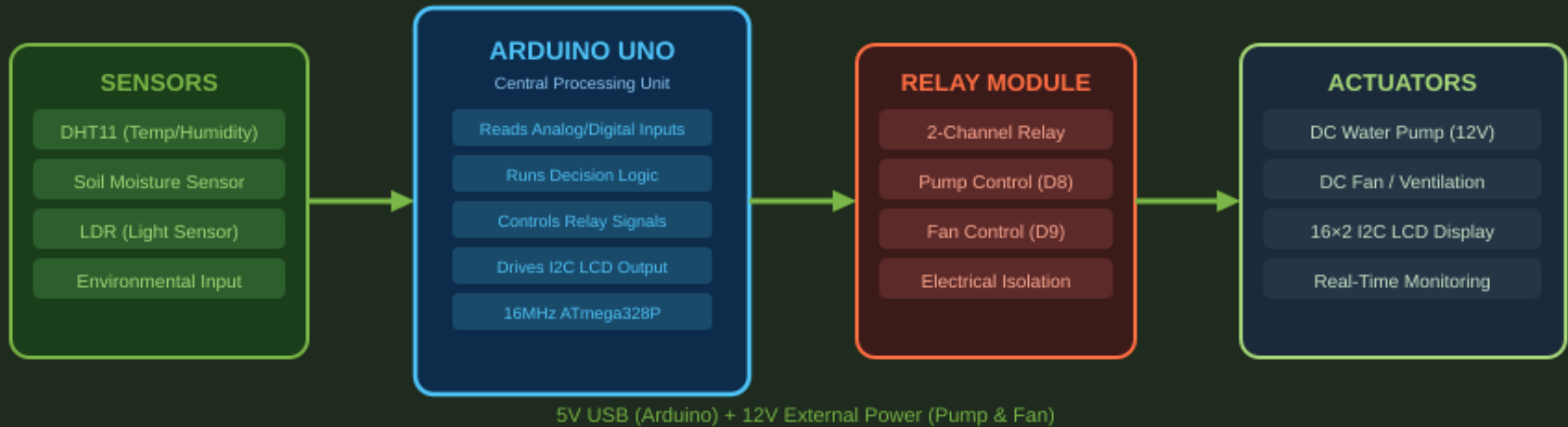
M-M, M-F connection wires



12V Adapter

External power for actuators

SYSTEM DESIGN & ARCHITECTURE



SENSE

Sensors collect temperature, humidity, soil moisture, and light data continuously from the greenhouse environment.

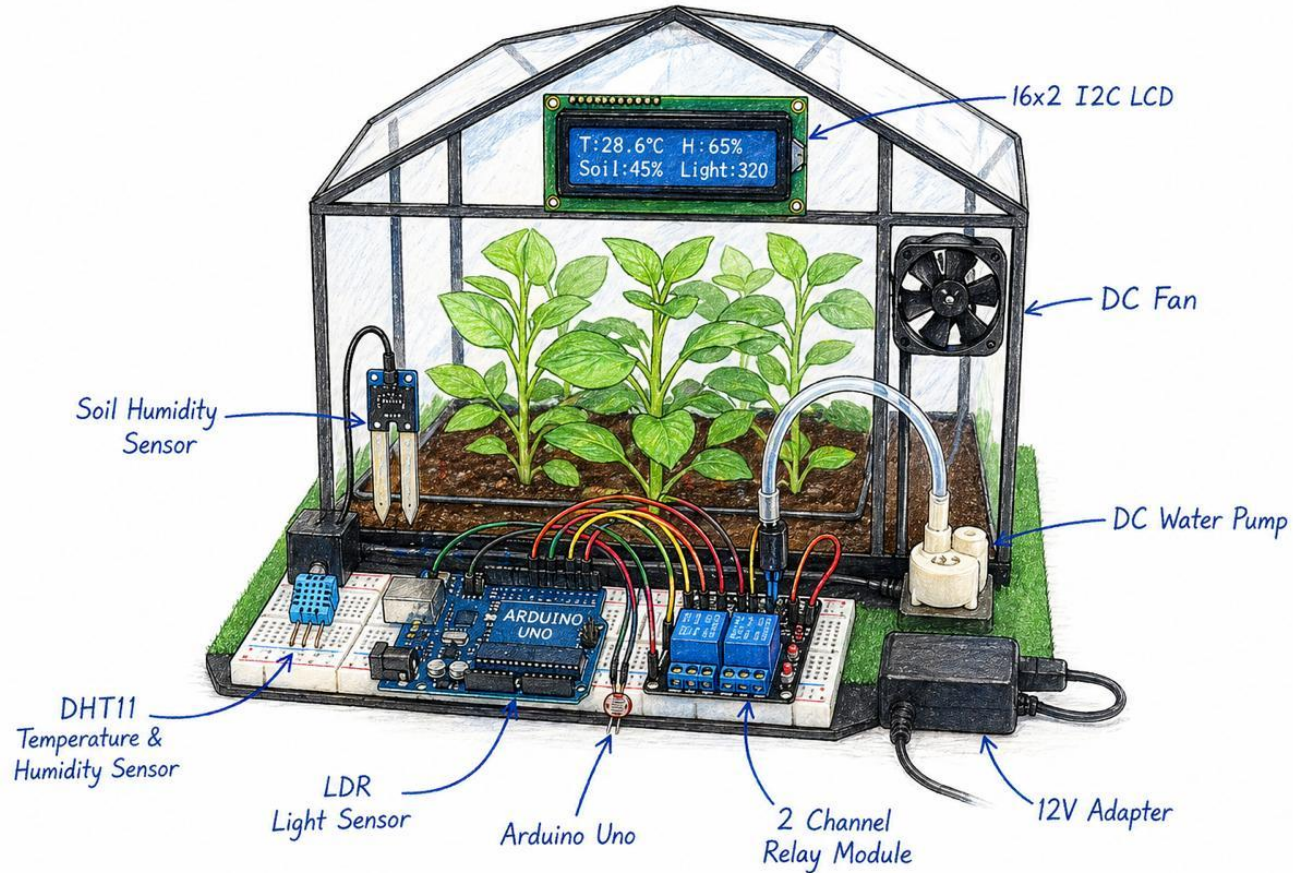
PROCESS

Arduino UNO reads analog and digital inputs, evaluates conditions using programmed thresholds, and generates control signals.

ACT

Relay module safely switches the 12V water pump and fan based on Arduino output. LCD displays live status to users.

PROJECT DESIGN



THE ARDUINO CODE

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>
#include <DHT.h>

#define DHTPIN 2
#define DHTTYPE DHT11

#define SOIL_PIN A0
#define LDR_PIN A1

#define PUMP_RELAY_PIN 8
#define FAN_RELAY_PIN 7

DHT dht(DHTPIN, DHTTYPE);
LiquidCrystal_I2C lcd(0x27, 16, 2);

int soilThreshold = 100;

// timing control
unsigned long lastPumpTime = 0;
unsigned long absorptionTime = 10000; // wait 10s after watering

void setup() {
  Serial.begin(9600);

  pinMode(PUMP_RELAY_PIN, OUTPUT);
  pinMode(FAN_RELAY_PIN, OUTPUT);

  // relays OFF (assuming FAN still active LOW)
  digitalWrite(PUMP_RELAY_PIN, LOW); // OFF (changed logic)
  digitalWrite(FAN_RELAY_PIN, HIGH); // OFF (unchanged)

  dht.begin();
  lcd.init();
  lcd.backlight();
}

void loop() {
  // 📡 Read sensors
  int soil = analogRead(SOIL_PIN);
  int light = analogRead(LDR_PIN);

  float temp = dht.readTemperature();
  float hum = dht.readHumidity();

  if (isnan(temp) || isnan(hum)) {
    temp = 0;
    hum = 0;
  }
```

```
void loop() {
  // 📡 Read sensors
  int soil = analogRead(SOIL_PIN);
  int light = analogRead(LDR_PIN);

  float temp = dht.readTemperature();
  float hum = dht.readHumidity();

  if (isnan(temp) || isnan(hum)) {
    temp = 0;
    hum = 0;
  }

  // 🌀 FAN CONTROL (unchanged logic)
  if (temp > 20) {
    digitalWrite(FAN_RELAY_PIN, HIGH); // OFF
  } else {
    digitalWrite(FAN_RELAY_PIN, LOW); // ON
  }

  unsigned long currentTime = millis();

  // 💧 PUMP CONTROL (FIXED ACTIVE HIGH LOGIC)
  if (soil < soilThreshold && (currentTime - lastPumpTime > absorptionTime)) {
    Serial.println("🚰 Soil DRY → Pump ON for 1s");

    digitalWrite(PUMP_RELAY_PIN, HIGH); // ON (FIXED)
    delay(1000); // 1 second pump
    digitalWrite(PUMP_RELAY_PIN, LOW); // OFF (FIXED)

    lastPumpTime = currentTime;
  }

  // 📺 LCD
  lcd.clear();

  lcd.setCursor(0, 0);
  lcd.print("T:");
  lcd.print(temp);
  lcd.print(" H:");
  lcd.print(hum);

  lcd.setCursor(0, 1);
  lcd.print("S:");
  lcd.print(soil);
  lcd.print(" L:");
  lcd.print(light);

  // 📡 SERIAL DEBUG
  Serial.print("Soil: ");
  Serial.print(soil);
  Serial.print(" | Temp: ");
  Serial.print(temp);
  Serial.print(" | Hum: ");
  Serial.print(hum);
  Serial.print(" | Light: ");
  Serial.println(light);

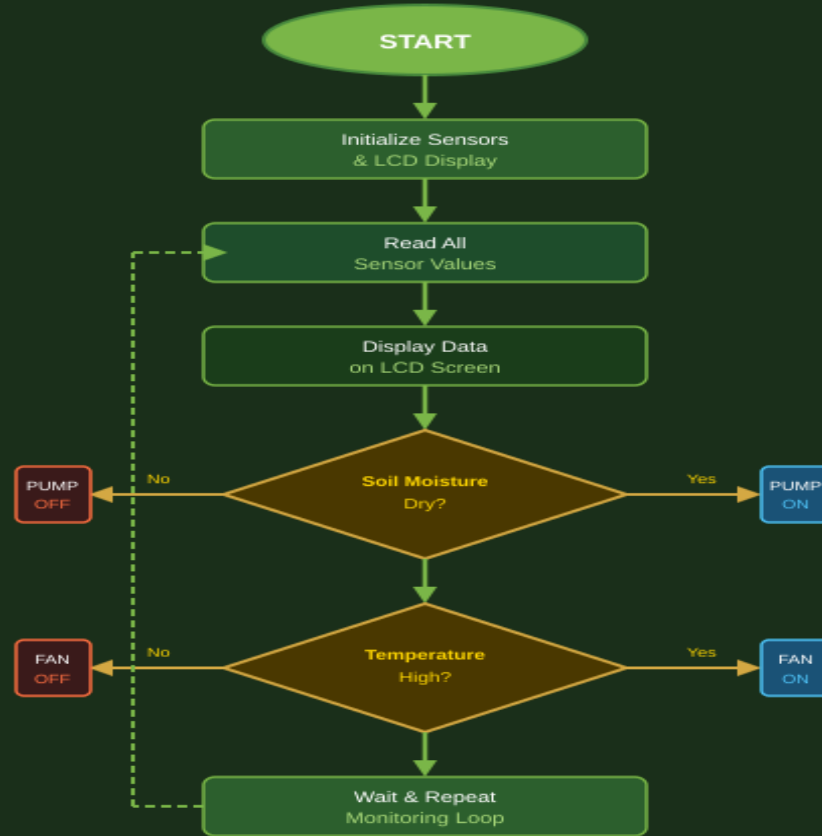
  delay(2000);
}
```

SYSTEM FLOWCHART

System boots and initializes all peripherals — sensors, relay, and LCD.

Analog & digital inputs sampled every cycle.

LCD updates every loop iteration.



Soil Moisture threshold: < 500 analog units activates pump.

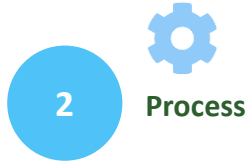
Temperature threshold: > 30°C activates ventilation fan.

SYSTEM OPERATION



Sense

DHT11, Soil Sensor & LDR continuously sample the greenhouse environment every second.



Process

Arduino reads all sensor inputs, compares to threshold values, and determines required actions.



Relay Switch

Relay channels D8 and D9 activate or deactivate based on Arduino's HIGH/LOW signals.



Irrigate

Water pump turns ON when soil moisture drops below threshold. Turns OFF when adequate.



Ventilate

DC fan activates when temperature exceeds 30°C, maintaining optimal growing conditions.



Display

LCD shows live T°, Humidity, Soil status, Pump state, and Fan state every cycle.

RESULTS & DEMONSTRATION

LCD Display Output

T: 30.60C H: 65%

L: 820 S: 2

Sensor Readings

30.60°C

Temperature

65%

Humidity

820 lux

Light (LDR)

☒ Irrigation Automated

Pump successfully activated when soil moisture < threshold. Plants watered without manual intervention.

☒ Fan Auto-Activation

Ventilation triggered at T > 30°C, maintaining optimal temperature inside greenhouse.

☒ Live LCD Monitoring

All sensor values displayed in real time. System state (Pump/Fan ON/OFF) clearly visible.

☒ Stable Monitoring Loop

System ran continuously without errors. All sensor readings consistent and reliable.

CHALLENGES FACED & SOLUTIONS



Relay Wiring Complexity

⚠ Problem:

Incorrect relay wiring caused Arduino resets and power instability when actuators switched on.

✓ Solution:

Added external 12V power supply for actuators and verified relay NO/NC/COM terminal assignments.



Pump Power Requirements

⚠ Problem:

DC water pump drew too much current for the Arduino's 5V pin, causing voltage drops.

✓ Solution:

Used relay to connect pump directly to 12V adapter, completely isolating it from Arduino power.



Sensor Calibration

⚠ Problem:

Soil moisture sensor readings varied across different soil types and moisture levels.

✓ Solution:

Calibrated threshold values by testing in dry/wet conditions and mapping to reliable analog ranges.



Breadboard Stability

⚠ Problem:

Loose breadboard connections caused intermittent sensor failures and erratic relay behavior.

✓ Solution:

Secured all wire connections, used shorter jumper wires, and tested continuity with multimeter.

CONCLUSION



Automation Achieved

System successfully automates irrigation and ventilation without manual intervention, 24/7.



Smart Agriculture Impact

Embedded systems significantly improve efficiency and sustainability of greenhouse farming.



Water & Energy Efficiency

Pump and fan only activate when needed, conserving resources and reducing energy consumption.



Low-Cost & Scalable

Built with under \$25 of components. Architecture can scale with additional sensors and actuators.

FUTURE IMPROVEMENTS

01



ESP32 Wi-Fi Integration

Replace Arduino Uno with ESP32 to enable wireless sensor data transmission and remote control over Wi-Fi.

02



Cloud Monitoring

Stream sensor data to cloud platforms (ThingSpeak, AWS IoT) for historical analysis and remote dashboards.

03



Mobile Application

Develop iOS/Android app for real-time monitoring, alert notifications, and manual override controls.



Solar Power Integration

Add photovoltaic panels and battery storage for fully off-grid, sustainable greenhouse operation.

05



AI-Based Plant Analysis

Integrate machine learning models to predict plant health, optimal watering schedules, and disease detection.

THANK YOU



Any Questions?

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